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Student identification of the social, economic and environmental implications of using Generative Artificial Intelligence (GenAI): identifying student ethical awareness of ChatGPT from a scaffolded multi-stage assessment

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ABSTRACT

Generative Artificial Intelligence (GenAI), typified by tools like ChatGPT, has generated widespread debate about its potential benefits and significant risks, resulting in confusion around social, economic, environmental, and ethical impacts. Instances of academic integrity breaches, undisclosed usage, corporate embarrassments, and intellectual property concerns have emerged. Little was known which of these implications students could identify. To gauge students' awareness, a scaffolded two-stage assessment was introduced. A 32-item catalogue of social, economic, and environmental implications was designed to identify those students could independently recognise, supplemented by a critical reflection process. In the first stage, students critically analysed ChatGPT outputs to identify ethical considerations. In the second, they conducted a more extensive research-based reflective exercise. Students identified all but three implications, but the average number identified was low. These findings show that students were more likely to recognise positive implications, emphasising the need to raise awareness of risks and ethical challenges.

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Assessment; ChatGPT; ethics; evaluative judgement; Generative Artificial Intelligence (GenAI)

Introduction

ChatGPT is a Generative Artificial Intelligence (GenAI) application that has required educational institutions and businesses from around the world to stand up and take notice by being the fastest-growing consumer application in history (Hu 2023). The technology has the potential to drive many transformational opportunities, including providing significant productivity gains across many industries, including enhanced throughput, efficiency, and quality (Bughin 2024). These identified gains in value creation must also be considered together with the risks (Kanbach et al. 2024). As pointed out by Daniel and Nikolic (2023) there are many usage risks and ethical considerations that the community needs to be aware of, including avoiding major career embarrassment, such as when lawyers included fictitious legal research in a court filing due to GenAI hallucination (Legg and McNamara 2024). Therefore, a balanced understanding of the positive and negative implications of its use is needed, which is a major focus of this study.

The education sector has not been immune to balancing the benefits and challenges associated with GenAI use (Crompton and Burke 2024). In early 2023, with the release of ChatGPT-3.5, the

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education sector quickly became concerned about how the technology could substantially threaten the integrity of unsupervised assessments and that changes to long-standing educational practices would be needed (Kizilcec et al. 2024). However, at the time, there were many limitations to its effectiveness. While ChatGPT-3.5 could pass many assessment types, for students to utilise the technology well, they needed to understand what a good/correct answer looked like, have good prompt engineering skills and awareness of its many drawbacks, including the tendency to hallucinate (Nikolic et al. 2023). The work of Nikolic et al. (2023) provided a strong benchmark of early GenAI capability against engineering assessments based on the multi-institutional and disciplinary approach. This study was conducted in a similar time period (Q1 2023) when GenAI literature was substantially lacking to provide a benchmark understanding of student knowledge regarding the possible implications of its use. This is important because the technology has evolved rapidly. A year later, ChatGPT-4 became substantially more capable than GPT-3.5, and the list of GenAI technologies grew (e.g. Copilot & Gemini), resulting in growing academic integrity risks (Nikolic et al. 2024). However, GenAI can also be seen positively as a tool enabling new opportunities to enhance teaching and learning (Farmera et al. 2023; Fatahi et al. 2023; Quince, Seligmann, and Maxwell 2024). With reports of ChatGPT-o3 beating human experts for the first time against the Graduate-Level Google-Proof Q&A test and Artificial General Intelligence claimed to be close to being achieved (Mollick 2025), the community needs to pick up the pace and understand how best to use it and live with it in a socially acceptable way.

Many schools and universities at first outright banned the use of GenAI, but students continued to use it (Purtill 2023), suggesting that there may be a significant gap in educating students and academics on how to use it ethically. Understanding ethical awareness is paramount because a raft of issues on its wrong use has already become a problem in the academic and corporate worlds, and we need to start educating students before the situation escalates (Daniel and Nikolic 2023). Therefore, ethics was decided to be a good starting point for engaging students in using GenAI. Ethics can help individuals and societies reflect on rights and wrongs and help provide guidelines on how to act and make decisions that promote fairness, justice, and overall well-being, and can connect to theories such as virtue ethics, care ethics, and the ethics of freedom (Grande et al. 2024). Due to the rapid rise of ChatGPT-3.5 use in early 2023, little was understood about the implications surrounding its use, including what initial implications students could identify. This was a major gap that this study tried to address.

Released before the GenAI surge of interest associated with the release of ChatGPT-3.5, Stahl (2021) provided great insights into the concept of ethics and its application to AI. Those insights highlight that ethical implications go well beyond academic integrity risks that gain much attention and cover a range of social, economic and environmental factors. Some of these factors are considered as concerns, while others are seen as benefits. Examples extracted from Stahl (2021) include: *Socially*, using GenAI raises significant concerns, such as threats to privacy, biases in AI decision-making, and the spread of misinformation, which can undermine trust and disrupt societal cohesion. However, AI also offers social benefits, including enhancing individual and societal well-being, increasing access to information, and improving everyday experiences through personalised education and healthcare applications. *Economically*, while GenAI can disrupt job markets by automating tasks, it also presents opportunities for innovation, increased efficiency, and new job creation in emerging fields. AI promises economic growth through improved productivity and can contribute to wealth creation and better standards of living. Yet, these benefits come with the risk of widening inequality, particularly if wealth distribution is not managed ethically. *Environmentally*, AI's energy-intensive nature poses a significant challenge due to its contribution to carbon emissions. However, AI also holds the potential for positive environmental impacts, such as optimising energy use, advancing climate research, and supporting sustainable practices through enhanced data analysis and predictive capabilities. Understanding these dual aspects of concern and hope is crucial from a student's perspective, as it is important to understand what they can identify.

While the work of Stahl (2021) provided strong insights into the ethical implications of AI, updated insights were needed that also considered the technology-specific implications associated with the generative capabilities. Without an available update, the authors expanded the work of

Stahl (2021) to develop 32 implications that considered both positive and negative implications. The hypothesis being that if students could consider these implications holistically, they could weigh up if, when, how, and how much GenAI should be used. The major focus of this study was to identify which implications students could identify on their own without being explicitly taught. An activity that could help students with the identification process and engage their critical thinking skills was developed and tested.

A purpose-built assessment was needed to determine students' capability to identify implications. The question became how to design a GenAI-integrated activity when there was minimal educational integration evidence at the time. All was not silent, and the education community has been working through the challenges of integrating GenAI use. For example, an organising framework developed by Bearman, Nieminen, and Ajjawi (2023) outlines how the community can purpose assessment design in a digital world. Applying the framework to GenAI, three assessment areas are proposed: 1. Using GenAI (legally and ethically) to support better assessment. 2. Assessing students on reading/writing/using GenAI, and 3. Assessing for uniquely human capabilities, which GenAI can't do. An important question arises: How does the education community transfer this framework into a practical implementation? There is a great need for implementation examples. These examples also need to be based on sound pedagogy. This is due to a gap discovered by González-Calatayud, Prendes-Espinosa, and Roig-Vila (2021), who found through a systematic review that most AI-based assessment studies focused on the technical aspects of incorporating AI rather than the pedagogical models underpinning their use in education. To address this, an assessment task based on research-informed pedagogical principles was designed and tested in this study.

A two-part assessment task was designed to encourage students to use higher-order cognitive skills to explore and develop their understanding of the ethical implications of using GenAI. It was a formative assessment designed for learning and extracting what students could identify themselves after exposure to broad digital ethics theory. This assessment could provide a benchmark understanding of student awareness, providing a pathway for pedagogical change and a connecting reference point for future studies. Therefore, this study attempts to answer the research question, *'Using an embedded multi-stage assessment, how many social, economic and environmental ethical implications can students identify?'* The results of this study will be valuable for anyone in the education community looking to understand the depth of students' initial GenAI ethical understanding (what they know before being explicitly taught). This study continues by synthesising the social, economic, and environmental implications and outlining the pedagogical approach before diving deep into the results, analysis, and discussion.

Determining the social, economic and environmental implications

One of the significant concerns surrounding the integration of AI and GenAI applications is the many unanswered questions surrounding the implications (Daniel and Nikolic 2023; Stahl 2021). How will users engage with the technology, and what are some ethical implications of such use? Even before the emergence of GenAI, there has been a need to increase the exposure of ethics to engineering students (Gwynne-Evans, Chetty, and Junaid 2021). Furthermore, as the engineering profession can have such a large impact on people's lives and society at large, there is a growing need to reinforce the human and social aspects of engineering practice, with recommendations for greater integration of human and social dimensions within the technical contexts (Martinez-Marroquin et al. 2024). Ethics is a complicated topic, and decisions can fall between what an engineer 'must do' and what 'would be good to do' (Chance et al. 2021) and they need guidance. Teaching staff are important in guiding and mentoring students in building their ethical perspective (Grande et al. 2024). This is especially critical in engineering education, where future professionals must address not only technical challenges but also the social, economic, and environmental impacts of their decisions.

Social, economic and environmental implications are three pillars of sustainability with no clear origin of definition but are important in addressing critical aspects of human and planetary well-

being (Purvis, Mao, and Robinson 2019). Sustainability has been taught in higher education since the 1970s, but the development of 17 sustainable development goals by the United Nations in 2015 to protect the interest of future generations and build universal peace has heightened its importance (Nakad, Gardelle, and Abboud 2024). The three pillars can be broadly defined as (Purvis, Mao, and Robinson 2019):

Social Pillar: This pillar focuses on equity, social justice, and improving quality of life, ensuring that human needs such as health, education, and cultural integrity are met.

Economic Pillar: Aims to maintain and enhance economic systems to support societal needs while emphasising efficient resource use and minimising environmental degradation.

Environmental Pillar: Ensures the sustainable use of natural resources and the protection of ecosystems, addressing issues like biodiversity, climate change, and pollution.

Teaching these pillars helps students understand the interconnected nature of global challenges, develop critical thinking about trade-offs, encourage innovative solutions, and prepare as future leaders to balance economic growth, social equity, and environmental preservation in decision-making. These considerations also need to be made in the context of GenAI. Varying opinions are present, with GenAI engagement being seen as an enabler of social, economic and environmental advancement (Wang et al. 2023), while others see a risk of greenwashing given the little evidence of life cycle assessment on its application (Wenmackers 2024). Therefore, students need to learn the social, economic and environmental ethical implications of GenAI at university to prepare them for responsible decision-making in their future careers. Understanding these implications equips them to navigate the complex challenges of privacy, fairness, sustainability, job displacement and inequality in an increasingly AI-driven world. By being aware of the impacts, students can contribute to the development and deployment of AI technologies that are both innovative and ethically sound, ensuring that technological advancements benefit society as a whole. It is about balancing the benefits of GenAI with the drawbacks. This is what makes this study unique, focussing on both to gain a more holistic understanding of the implications.

When data analysis commenced, no systematic literature review of such GenAI-based ethical implications could be found. This was needed to help determine a holistic set of implications based on the Three Pillars of Sustainability – social, economic and environmental (Purvis, Mao, and Robinson 2019) against which the student work could be benchmarked. To overcome this, the authors started exploring grey and scholarly literature to identify as many as possible. Due to the novelty, this list was developed for research purposes after students had completed the assessment, providing an opportunity for more scholarly literature on this topic to be published. Most studies repeated common ethical implications centred around bias, privacy and integrity. Furthermore, most concentrated on ethical concerns rather than benefits. From this search, the work of Stahl (2021) was considered the most comprehensive. While focussed on AI rather than GenAI, the limitations were transferable. Beyond this, a select number of studies (Daniel and Nikolic 2023; Fosso Wamba et al. 2023; Lindsay, Johri, and Bjerva 2023; Shi et al. 2023) presented some additional possible implications. After synthesising the implications from these papers, 32 were identified and are expressed in Table 1. Each implication has been assigned a code from 1 to 32, which is used for the analysis.

A limitation of this synthesis approach was that it may have missed some possible items or only focused on a singular viewpoint compared to a systematic literature review (SLR). Due to the need to gather and share such insights, the authors believed that this approach would, at the least, provide a starting point for others to build upon. To further overcome this limitation, in parallel to this study, the authors set up a separate team to conduct an SLR on the ethical implications within engineering education and practice (Quince, Petkoff et al., 2024). Due to the focus on journals and the scope of the SLR, no improvement was made to the 32 codes identified in this study. Additionally, prior to submission, the authors checked for other newly published SLRs to confirm if anything had been missed as a limitation. One of the most comprehensive newly published studies that focussed on

Table 1. A list of possible GenAI-based social, economic & environmental ethical implications and uses.

Code	Implication	Brief explanation
C1	Affect recognition	Affect recognition interprets emotions via facial expressions, body language, and speech, offering potential benefits in areas like mental health support, adaptive learning, and customer service. However, it faces criticism for methodological flaws and inconsistency. Privacy concerns arise with its use in surveillance, like in schools, and it can perpetuate bias, as seen with an algorithm leading to racial profiling by linking behaviours to 'terrorist behaviour'.
C2	Data augmentation	GenAI serves as a tool for data augmentation, expanding data quantity and diversity, and is usable to train new AI models. Helping AI systems learn better can lead to economic advantages, but its ethical implications depend on its application. It can lead to environmental strain, social harm, and economic disruption if not carefully managed.
C3	Process automation	GenAI offers versatile and powerful capabilities to streamline and optimise tasks across various domains, transforming how processes are automated. It may lead to major gains in efficiency and safety, but it comes with risks including widening inequality, displacing jobs, and stressing the environment if not implemented responsibly.
C4	Bias	GenAI is a vast language model trained on a large dataset like common crawl, which gathers data from web pages and other sources over many years. It grants immense capabilities but also harbours biases. Internet scraping can introduce discriminatory language alongside valuable content.
C5	Collaboration	GenAI can enhance collaboration by facilitating idea exchange, streamlining communication, and fostering shared insights for collective problem-solving. The impact depends on how it is used. Either as a helpful tool that supports people or as something that undermines human contribution.
C6	Competitive advantage	Gen-AI tools can facilitate competitive advantage in the face of rapid digital transformation. This may include improved resource allocation, capability development, efficiency and increased revenue/profit through cost reduction and improved productivity. However, this can also widen the gap between organisations with access to advanced AI and those without.
C7	Solution exploration	Users can use GenAI to explore potential solutions or lines of inquiry when presented with a challenge or question. How fair, helpful, or sustainable those solutions are depends on how we use the technology.
C8	Control and oversight	This relates to ensuring that humans remain in ultimate control of all critical processes. AI should be used to support, not replace, human judgment, especially in decisions that affect people's lives, rights, or well-being.
C9	Copyright and intellectual property	Linked to ethical integrity are concerns about intellectual property and copyright. For example, AI-generated image tools like Stable Diffusion and DALL-E use internet images, sparking debates on artists' rights as their styles get replicated without consent. Some argue this mirrors traditional artistic processes, while artists view it as an ethical breach.
C10	Data collection and utilisation	The process of collecting data, transforming it (datafication), and leveraging it (big data). This may lead to smarter technology that can unlock innovation, improve services, and support economic development if done appropriately. Datafication transforms nearly all areas of our daily lives into data for AI algorithms, raising serious privacy concerns. From location to health data, our interactions feed algorithms, turning user data into products and can lead to exploitation. Big Data can exacerbate biases and discrimination, as the algorithm's perspective lacks diverse input due to unequal access to technology.
C11	Data ownership	This represents the concerns about who owns the data generated by GenAI and uploaded to GenAI.
C12	Data security	Security and protection of data are needed as there are vulnerabilities and defences in LLMs that need to be considered. This includes AI Model Inherent Vulnerabilities (e.g. data poisoning, backdoor attacks, training data extraction) and Non-AI Model Inherent Vulnerabilities (e.g. remote code execution, prompt injection, side channels).
C13	Decision making, risk & uncertainty	GenAI may shape human decision-making by influencing how risk and uncertainty are perceived and managed. While it can support faster, data-informed choices, care is needed to avoid superficial or contextually inaccurate responses that may lead to poor or overconfident decisions.
C14	Enhanced user experiences	GenAI's ability to generate content and adapt to user needs may enhance experiences across platforms—improving quality through visuals, language, and communication efficiency. However, there are risks of over-personalisation, reduced human agency, and user manipulation if experiences are optimised solely for engagement or profit.
C15	Environmental impact	The lifecycle of GenAI can contribute to greenhouse gas emissions due to its reliance on energy-intensive data centres and cloud computing. While less direct, the mining and processing of materials used in AI hardware can also lead to environmental harm.
C16	Equity and accessibility	Fair and unbiased access to the benefits of GenAI for all users, regardless of background, resources, or abilities. This includes making the technology available and usable by diverse communities while mitigating biases and ensuring equal opportunities for all.

(Continued)

Table 1. Continued.

Code	Implication	Brief explanation
C17	Explore alternatives	GenAI draws on diverse human knowledge to generate a wide range of outputs across text, visuals, and more. This enables users to engage with abstract concepts in varied and accessible ways, encouraging the exploration of alternative ideas and perspectives. For example, a user might prompt GenAI to generate counterarguments to their own viewpoint, fostering critical thinking and deeper understanding. However, GenAI may also present all perspectives, regardless of accuracy or credibility, as equally valid. This can lead to confusion, reinforce existing biases, or unintentionally spread misinformation, particularly if users lack the context to evaluate the responses critically.
C18	Feedback & improvement	GenAI can provide diverse feedback types, including performance reviews, learning insights, actionable suggestions, and improvement strategies. However, its feedback may lack accuracy, context, emotional intelligence, or an understanding of nuanced human experiences, leading to generic, inappropriate, or even demotivating responses if not carefully interpreted.
C19	Human labour	GenAI's impact on human labour is complex. While it creates efficiency and new job opportunities, it also threatens to displace workers, particularly in low-skill or repetitive roles. Even in high-skilled fields like law or finance, automation raises concerns about job security. Early GenAI systems often relied on hidden, underpaid human labour for tasks like data labelling and moderation. Ethical deployment must consider both the risks of exploitation and the potential to enhance work if guided by fair policies and inclusive design.
C20	Ethical Integrity	Users may use language models such as ChatGPT to declare outputs that are not their own. Note that GenAI output is not traditionally-defined plagiarism, as the model's output is not copied from a single source. Rather, the output is an original creation generated 'probabilistically' from a wide range of sources.
C21	Learning	GenAI-powered curriculum design can produce personalised and engaging experiences, for learning and skill development. It could cause over-dependence and a decline in skills such as critical and independent thinking.
C22	Organisational structure	GenAI can be integrated into organisational structures and roles, providing the potential for innovation, while enhancing efficiency, responsiveness, and agility. However, its adoption may also lead to challenges such as increased centralisation of decision-making, reduced transparency, and over-reliance on AI-driven processes. These shifts can undermine human oversight, disrupt team dynamics, and create resistance or confusion among employees if not managed carefully.
C23	Performance measurement	GenAI can refine performance measurement by automating data analysis, offering real-time insights, and optimising metrics for accurate and informed assessments. It can create performance cultures focussed on metrics rather than intangible higher values of contribution and service.
C24	Power and hegemony	GenAI models, built on static data, reflect the existing societal power dynamics and their limitations and biases, potentially further marginalising disadvantaged groups. These models encode a dominant perspective, often reflecting a heterosexual, white, Western male lens due to the volume of internet content. Efforts to create fair synthetic data still reproduce biases. Additionally, AI reinforces global power imbalances, as access to resources and wealth concentrates AI in the hands of the already powerful, deepening divides between wealthy and poorer nations and people.
C25	Privacy	Concerns include personal data usage, breaches, and opaque decision-making. Issues like lack of consent, unclear data ownership, and biased profiling raise serious risks especially in applications like facial recognition, where privacy and civil rights may be compromised.
C26	Psychological well-being	GenAI systems can shape users' mental and emotional states through their design and interactions. While they may support well-being by offering guidance, reducing stress, or providing emotional support, they can also pose risks such as increasing anxiety, encouraging over-reliance, or exposing users to manipulative or harmful content. Ensuring psychologically safe and ethical design is essential to protect users' mental health.
C27	Output quality and accuracy	Determining if the output of the GenAI is reliable, relevant, and factually accurate. Evaluating quality requires evaluative judgment, as outputs must reflect real information, not fabricated or "hallucinated" content, and align with current, verified knowledge. Misinformation can have serious consequences.
C28	Refine outcomes	Users employ GenAI applications with specific goals to refine and generate outputs that meet their qualitative or quantitative requirements. While fine-tuning can help improve specific goals, such as the quality of content and meeting certain criteria or performance-based requirements, over-refinement can unintentionally reinforce user biases, rely on flawed inputs, or give a misleading sense of accuracy.

(Continued)

Table 1. Continued.

Code	Implication	Brief explanation
C29	Safeguarding against misuse	Measures to prevent unauthorised access, manipulation, or malicious use of AI systems. The rise of using 'deep-fakes' for fake news is one example.
C30	Transparency and accountability	The responsibility held by developers, users, and stakeholders for the GenAI's actions and outcomes. It involves transparency in how AI systems operate, ensuring ethical use, addressing biases or errors, and being answerable for the impact and decisions made using AI technology.
C31	Improved understanding	GenAI can enhance users' comprehension across subject domains by providing explanations, examples, and diverse perspectives that may support deeper learning. It may help users grasp complex concepts and explore new ideas more efficiently. However, there is a risk of oversimplification, misinformation, or users developing surface-level understanding if they rely too heavily on AI without critical thinking or cross-checking with reliable sources.
C32	Language fluency	GenAI may enhance language fluency by improving grammar, vocabulary, and expression. However, over-reliance may weaken independent language skills and reduce mother tongue fluency over time.

ethical concerns by Bukar et al. (2024) was identified but contained no new limitations. This suggests that this list is the most comprehensive to date and a valuable resource for the community to build upon, presenting a major contribution to the field.

Within the review process, the 32 items were tested against ChatGPT-4o (OpenAI 2025) for completeness to gain a GenAI perspective. It provided 8 possible missing implications. This included cultural, democratic processes, mental health, long-term economic disparities, ethical development of GenAI models, regulatory and legal challenges, education and skill development gaps and epistemological impacts. However, the authors do not believe adding them to the list would bring much value. The added implications could be considered subsets or extensions of the original 32 items, as concerns like cultural impacts, regulatory challenges, and epistemological shifts can be encompassed within broader categories like bias (C4), transparency and accountability (C30), psychological well-being (C26), power and hegemony (C24) and learning (C21). By maintaining focus on these higher-level, more generalisable items, the original list ensures clarity and avoids over-complication while still allowing room for nuanced discussion within each category. The final list of 32 ethical implications can be viewed in Table 1.

The 32 implications are not engineering specific, and they are important for every student engaging with GenAI as a tool to become aware of. However, the context and importance of each item will vary across disciplines. From an engineering perspective, some of the most important social, economic and environmental implications include:

Bias (C4), Equity and Accessibility (C16), and Power and Hegemony (C24): Engineers must ensure that GenAI models do not perpetuate societal inequities or existing power imbalances. This is crucial when developing safety-critical systems or infrastructure in contexts with diverse user needs. These issues are particularly pressing in global applications of AI, where unequal access to resources can widen technological and economic disparities.

Environmental Impact (C15): Engineers are uniquely positioned to address the energy consumption and carbon emissions associated with GenAI. They can explore innovations in energy-efficient hardware and sustainable data centres to mitigate these impacts while leveraging GenAI for advancements in climate research and sustainability practices.

Privacy (C25) and Data Security (C12): The risks associated with GenAI's capacity for extensive *data collection and utilisation (C10)* require engineers to design systems that prioritise user privacy and incorporate robust cybersecurity measures. These skills are essential in sectors like healthcare and financial services, where breaches can have severe consequences.

Control and Oversight (C8) and Transparency and Accountability (C30): In engineering, maintaining human oversight of GenAI systems is essential, particularly in safety-critical domains such as transportation and energy. Transparency in AI decision-making processes is necessary to build public trust and ensure accountability in failure cases.

Human Labour (C19) and Competitive Advantage (C6): GenAI’s ability to *automate processes (C3)* has dual implications: it can increase efficiency and innovation but also risks displacing workers. Engineers must design systems that complement human expertise rather than replace it, fostering opportunities for upskilling and creating new roles.

Collaboration (C5) and Enhance User Experiences (C14): GenAI’s ability to facilitate idea sharing and improve user experiences is particularly relevant to engineering teams working on multidisciplinary projects. These capabilities enable engineers to approach complex challenges with greater creativity and efficiency.

Feedback and Improvement (C18) and Learning (C21): GenAI can support lifelong learning for engineers by providing tailored feedback and promoting skill development. This aligns with the engineering profession’s emphasis on continuous professional growth in response to technological change.

Ethical Integrity (C20): The issue of academic and professional integrity is central to engineering education. Engineers must critically assess the outputs of GenAI to avoid unethical use and ensure the reliability of GenAI-assisted designs and decisions.

The pedagogical approach

As the implications the study was investigating were determined, the assessment design and the pedagogical approach followed. The learning outcomes for the designed activity were for students to become familiar with (1) prompt engineering and ChatGPT’s output, (2) assess quality, (3) undertake research, and (4) identify the assessment design’s inbuilt explicit and implicit critical thinking elements. The core outcome was for students to develop a greater awareness of the social, economic and environmental ethical implications of GenAI use. The assessment was designed around principles: developing evaluative judgement, research, critical thinking and feedback. Each principle was implemented distinctly and addressed separately within the approach. Part A focused on developing evaluation judgement by critically analysing a ChatGPT output, and Part B focused on the feedback principle through a reflection and research activity. An overview of the approach is shown in [Figure 1](#). Parts A and B were conducted as part of the same assessment item in an unsupervised environment, submitted together.

Key insights into the learning outcomes include:

- (1) Prompt engineering: The students were given introductory insights before the assessment on prompt engineering. Part A allowed students to test their understanding by identifying that a basic prompt would only provide basic information. Students were directed to consider better prompts as part of the activity to aid in evaluating the given output. This demonstrated that if GenAI was used as a tool, it could not be used blindly.
- (2) Assess Quality (the context-specific definition of ‘what good looks like’): In Part A, students were guided to identify missing implications (this included deciding how many missing implications were sufficient), identify the reliability of the information (e.g. checking for hallucinations, which was common with ChatGPT-3.5 and the quality of the information sources) and comment on the writing style (identify the unnatural language that plagued ChatGPT-3.5 early on).

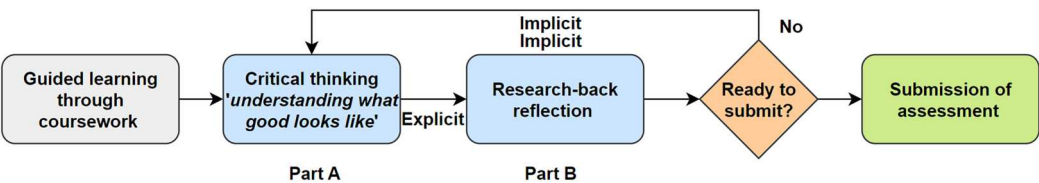


Figure 1. The pedagogical structure. The blue boxes represent the assessment components within the system. Parts A and B are submitted together.

- (3) Undertake Research: In Part B, students were guided to support their insights with references from traditional literature sources. This encouraged them to undertake research and critically engage with quality traditional sources.
- (4) Explicit and Implicit Critical Thinking: This tested the student's ability to think about the activities they were undertaking and connect the process to improve their learning. This was aided by the design that students were learning by doing. This included making judgments about what they were learning, the tools they had access to, and the decisions that they needed to make.

The assessment was the first that the students faced in the course and would provide an underpinning of all following assessments throughout the course. As such, it was designed to provide feedback to students and instructors during the learning process. Students were given insights into the ethical implications of technology in engineering through the coursework. This assessment was designed to help the students connect the theory with the practice, gain a foundational understanding of GenAI, and engage with the implications in much greater depth. The goal was for them to learn the implications by doing. For the teaching team, it provided a benchmark of which GenAI ethical implications students could identify, and how GenAI could be integrated. This implementation was conducted at a time when the exploration of GenAI integration was a key focus. This was because strong pedagogical know-how of successfully integrating GenAI into classrooms was not known. Literature on GenAI in January 2023 was limited (Nikolic et al. 2023). It is noted that by the time this paper completes the review process, our understanding and insights will be substantially stronger, and GenAI's capability will be much higher.

Assessment design principle: developing evaluative judgment

The design of the assessment task is aligned with the work of Bearman et al. (2023), focusing on two enabling principles: the first, the importance of designing feedback sequences to support learning and the second, to develop student capability to identify 'what good looks like' as was described earlier. The importance of the need for students to evaluate the output quality of GenAI is well documented. In fact, as Nikolic et al. (2023) discovered, identifying 'what good looks like' was paramount to using GenAI successfully for assessments that required higher-order cognitive skills. Mollick and Mollick (2022) proposed numerous 'understanding what good looks like' styles of assessment to be used with GenAI. They believed such implementations could overcome three classroom learning barriers: improving transfer, breaking the illusion of explanatory depth, and training students to evaluate explanations critically. Scaffolding upon this work, Vasconcelos and Santos (2023) implemented an 'objects-to-think-with' assessment that they claim supported students to hone their problem-solving, critical thinking, and creative skills. While these studies suggest that such an approach is highly successful in developing and confirming higher-order competencies, the data set is limited, and more evidence is needed. Bearman et al. (2024) defined this as helping students develop good 'evaluative judgement'. Therefore, this became central to the assessment design. This would require higher-order cognitive thinking skills from the students.

Focusing on higher-order cognitive skills is important to move away from GenAI's foundational application of helping students recall knowledge to provide answers to questions. When this is considered within the framework of Bloom's revised Taxonomy (Krathwohl 2002), the ability to remember (recall facts) is foundational, providing a scaffold to higher-order cognitive skills of analysing, evaluating and creating. These higher-order skills are core to Level 7 and higher of Australia's Quality Framework for the level of bachelor's degree and above (Australian Government 2016).

Assessment design principle: feedback

Most AI assessment implementations conducted to date have focussed on using formative assessment to guide feedback (González-Calatayud, Prendes-Espinosa, and Roig-Vila 2021). Generative

Artificial Intelligence provides an extraordinary opportunity to increase the frequency, quality and variety of feedback given to students, but with this comes a range of ethical concerns that must be considered (Lindsay, Johri, and Bjerva 2023).

Feedback plays a central role in the learning process (Black and Wiliam 1998; Nikolic, Stirling, and Ros 2018; Simpson-Smith 2024) and plays a major role in pedagogical decisions. Many forms and terms are used for feedback: informal, formal, formative, summative, student peer, student self, and constructive feedback. Not all feedback leads to performance improvement, with successful traits of effective feedback linked to being frequent, detailed, focusing on performance and learning, timely for application, aligning with assignment aims and learning concepts, attended to, and acted upon (Gielen et al. 2010). Even the best-intentioned feedback designs can do little for learning if the student decides not to act and reflect upon the feedback (Nikolic, Stirling, and Ros 2018).

Considering the overarching goal of developing students' higher-order cognitive critical thinking skills, where analysis and evaluation are central, a self-assessment/feedback approach was proposed. This is because students are always self-assessing, albeit informally and subconsciously, and it is important to ensure that they develop good judgment as a key outcome of their education (Taras and Wong 2023). The reported limitation of feedback based on self-assessment is that students struggle to identify their weaknesses without feedback from the teaching staff (Taras 2003). Given that, in real-world GenAI usage, individuals won't generally have access to someone who will tell them when the GenAI output is wrong, such a skillset needs to be developed. Therefore, a scaffolding activity, Part B, was designed to help students identify limitations in their capability and work and better recognise and develop the skills to determine 'what good looks like'.

Methodology

Evaluative judgment – assessment Part A

Part A of the assessment task was for students to use their evaluative judgment by providing feedback on an output generated by the subject coordinator using ChatGPT. The goal was to test if students could identify 'what good looks like'. The output was based on a prompt purposely designed with limited context to exhibit a lack of depth. The output was based on the prompt '*Write a 600-word essay on the social, economic and environmental implications of ChatGPT using Harvard style referencing*'. Beyond identifying the implications, students were to consider if the response was complete, reliable, accurate, and had a high-quality output in terms of how it was written and referenced. They then needed to consider how they could improve the prompt to develop a quality output further. This activity would target the higher-order cognitive skills of analysing, evaluating and creating, as outlined in Bloom's revised Taxonomy (Krathwohl 2002). Students would achieve this by analysing the information provided by ChatGPT, synthesising and verifying it against other information sources, and evaluating its accuracy and currency. Then, they could use this information to create a better prompt that could generate an output that would identify further ethical implications. To help students establish 'what good looks like' Part B was explicitly connected to Part A, designed to provide appropriate feedback to make such an evaluation.

Feedback – assessment Part B and the critical thinking and feedback cycle

In Part B students needed to find the positive and negative implications of GenAI use in their discipline, study life and future. They did this through reflection and research via an essay. By connecting this activity with research, the hope was that students would gain greater awareness of the diverse possibilities of implications and connect this newly gained knowledge with the work they had completed in Part A. This is because Parts A and B were submitted together for grading, connected together explicitly. By identifying missed implications, it was hoped that students would identify the weaknesses of their work in Part A on their own without first needing explicit feedback from the teaching staff. This

should have provided informal feedback on their evaluative judgment capability. More importantly, how, through reflection and research, they can better understand ‘what good should look like’. These research and reflection skills further extend the higher-order cognitive skills targeted in Part A.

In Part A, students were explicitly asked if ‘anything was missing’. When undertaking the research in Part B, students should have realised that there were implications that they could not identify on their own in Part A and updated their responses as appropriate. This was designed as implicit feedback. It tested student’s critical thinking capability to loop back and update their work in Part A.

At the minimum, Part A should have helped students identify the same or more implications found in Part B. The design also included an implicit feedback cycle. It was not explicitly instructed, but the researchers wanted to know if students could use their critical thinking skills to identify this implicit opportunity to reflect further and improve their work. It was unknown if the students would build upon what they learned in Part B to improve their evaluation of the output quality in Part A. That is, while the learnings of Part A add to the reflective and research learning process of Part B, can the students then see the implicit connection of using the learning from Part B to go back and improve Part A? This process is demonstrated in [Figure 1](#). This process requires students to make judgements on whether both parts are ready to be submitted, with the goal of helping students to develop good evaluative judgement.

The underlying focus of the learning and assessment process was for the students to understand the social, economic, and environmental implications and use cases when engaging with GenAI. To accomplish this, understanding these implications needs to be better understood.

This study was conducted in January-February of 2023 at the University of Southern Queensland (UniSQ). Due to academic integrity risks, the university did not allow students to use any GenAI at the time. The course focuses on the technology, sustainability and society of engineering and surveying-related disciplines. At the time, ChatGPT was the most popular and talked about GenAI application. ChatGPT-3.5 was also free and well-known for hallucinations and poor research output (Nikolic et al. 2023), making it ideal for Part A of the learning activity. The cohort size was 64 students, and it was a mixed cohort ranging from first to final-year students from all engineering disciplines taught at the university. As ChatGPT was reasonably new at that time, all students would have had limited prior access to the technology. The context of this timeframe must be well understood. Students had very low reported levels of AI competency compared to other digital technologies (Nikolic and Beckman 2025). What the technology is now capable of (Mollick 2025) and what the academic community now understands about the technology is more advanced (Nikolic et al. 2024). With this context, this study provides important benchmarking insights.

This study was conducted with ethics approval (ETH2024-0388). The approval conditions required all assessments to be de-identified to remove any bias or privacy concerns. This meant it was not possible to check for effects at the student year level (assuming students in higher years should have formed better evaluative judgment skills). Additionally, grades and feedback were removed before analysis as part of the de-identification process. The only possible analysis that aligned with the research question was developing insights into which social, economic, and environmental ethical implications of GenAI use students could identify. Therefore, the quality of the submissions was not a criterion.

Using the holistic set of implications identified in [Table 1](#), the researchers reviewed Parts A and B of each submission and, where applicable, matched the student response to assigned codes. A tally of each code was developed collectively for parts A and B, and the identification was recorded per submission. This was undertaken for quantification purposes. From this, the following possibilities were considered:

Possibility 1: The students would not recognise the explicit connection between Parts A and B, and different implications would be identified. Therefore, Part B, due to its research component, would contain a greater number of implications than Part A.

Possibility 2: The students identified the explicit and implicit connection between Parts A and B, and therefore, many of the same implications would be identified.

From this analysis, we can develop an understanding of:

- (1) Students' ability to research and reflect beyond the given example in Part A
- (2) The likelihood of students to recognise the implicit critical thinking loop
- (3) Determine which implications are most prevalent in students' thinking
- (4) Determine which implications need greater awareness by students
- (5) Reflect on the implementation of this pedagogical approach, including the ethical considerations from the perspective of the students

Limitations

This study was conducted at only one university and one sample of students. The sample size of 64 students is relatively small and can affect the generalisability of the study's findings to a broader population. Without a guiding systematic literature review, the full spectrum of implications was not known at the initiation of analysis, but as outlined earlier, it has been checked against recently published studies. Due to this study being conducted in Q1 2023 when little was known about ChatGPT, this study does provide an important benchmark and starting point for the conversations that need to be had and the research that needs to be undertaken. Due to the extra requirements to gain ethics approval to include focus groups better to understand students' engagement with the pedagogical design, this was excluded as it would not have been possible to undertake data collection on time. Due to the de-identification process we cannot measure for effects on year level or mark obtained. Unsupervised essays have now been identified as a high GenAI-based assessment integrity risk allowing students to bypass the reflection and research activities with little understanding (Nikolic et al. 2024). As the assessment was conducted early, towards the release of ChatGPT-3.5 those risks were low, and none of the submissions provided any indication that they were GenAI generated. If this study were to be repeated today, the confidence of the data would be much lower as it would be more likely that students would use it extensively to gain greater insights, and it would be impossible to detect how it was used.

Results & discussion

Figure 2 represents the number of implications students identified in Part A and Part B. It shows that students were able to identify more implications in Part B. For example, nine students were only able to find one implication in Part A, but in Part B, only two students found one. The ramifications of this are discussed in the following sub-sections.

The social, economic, environmental & ethical implications identified by students

For Part A, the critical review of GenAI output, 62 of the 64 students submitted the assessment task. Students, on average, identified 2.69 implications, and the median was 2. One outlier student identified 13 implications, less than half of all possible outcomes. Ten students were unable to identify any of the implications. Additionally, 71% of students could only identify three or fewer implications. This low percentage indicates that students struggled to evaluate the GenAI output directly via this activity, correlating with advice suggesting that teaching students to analyse GenAI output critically is an important learning activity to improve such competency (Ansari, Ahmad, and Bhutta 2024; Bearman et al. 2023). The low identification rate suggests that the diverse range of implications is not obvious to all and that students need guidance to understand the possibilities.

For Part B, 63 of the 64 students submitted the assessment task. Students, on average, identified 5.63 implications, and the median was 5. This was a substantial improvement compared to Part A, confirming our predicted possibility 1. This data indicates that a research activity was needed to

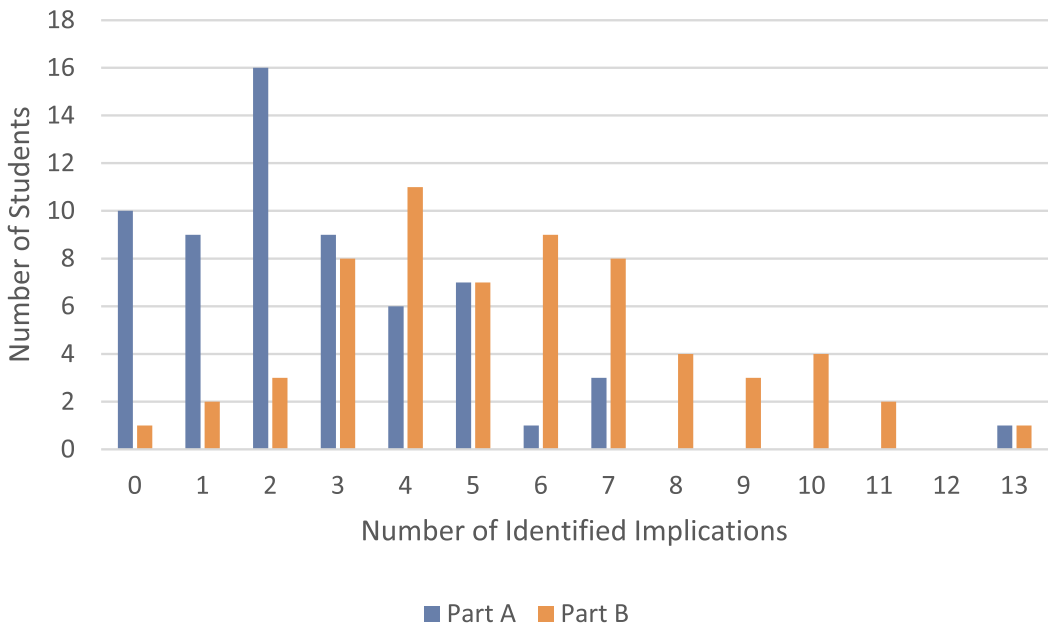


Figure 2. The number of identified implications identified by students in Part A and Part B.

help students become more aware of the ethical implications compared to simply making evaluative judgments on some text. It highlights that it shouldn't be assumed that students know the ethical implications on their own without educational scaffolding. This could include initial engagement with how GenAI technologies work or structured activities such as through tools like the Digital Ethics Canvas (Hardebolle et al. 2023). This highlights the need for GenAI ethics awareness in the curriculum. As research activities can help students manage unpredictable and uncertain future-facing challenges (Brew and Saunders 2020), the benefits of incorporating a research-based activity into the scaffold have been observed. As with Part A, 13 was the highest number of implications identified. By including Part B in the design, only 22% of students (down from 71% in Part A) could identify only three or fewer implications. The students did not identify any implication outside the 32 used in this study. It is important to acknowledge that students were not shown the list of implications before attempting the assessment.

Application of implicit feedback cycle

As demonstrated in Figure 1, an implicit feedback loop was pedagogically designed to test if students could determine the connection between the learning activities. As designed, the implications found in Part A (critical analysis of ChatGPT text) should have provided direction for Part B (research component). Then, Part B should have provided students with the capability to loop back and improve the analysis undertaken in Part A. Doing so would have strengthened the submission for both Parts A and B, indicating that the students understood both components were now ready to be submitted, and would have demonstrated students' ability to provide good evaluative judgment.

Determining the exact number of students who took advantage of this implicit feedback loop is impossible. However, one marker of this occurring is identifying the number of students that outlined the same implications for both assessment tasks. That is, if students applied this technique, students should have mostly raised the same implications for both parts of the assessment. Only 60 students submitted both parts of the assessment. The findings show that there was little crossover in the identification of implications across Parts A and B. Instead, the students mostly identified

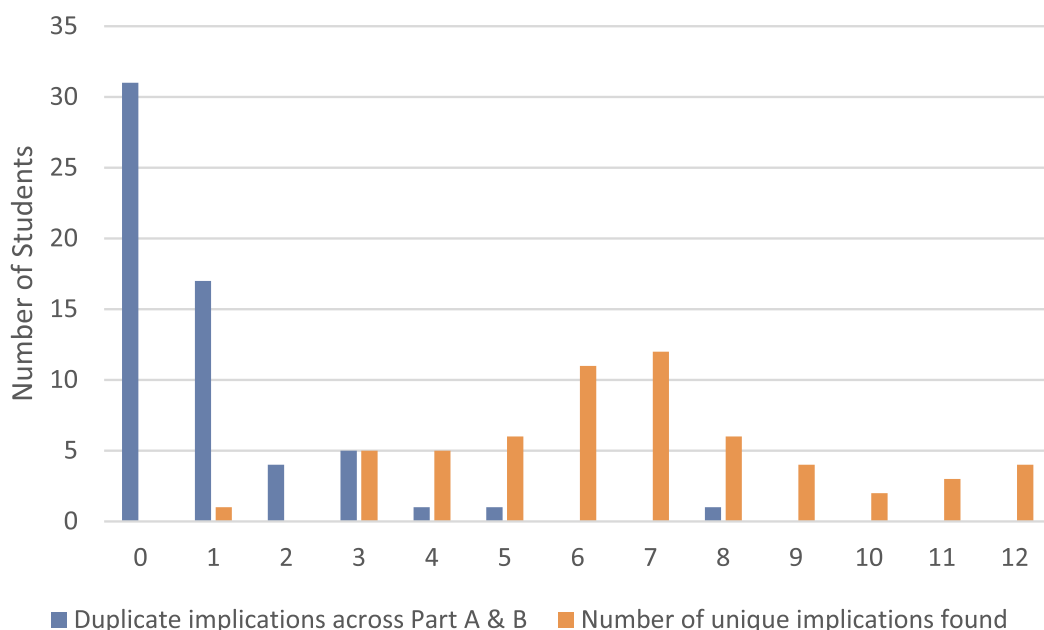


Figure 3. The total number of duplications identified by students (blue) and the total number of implications found across both assessment tasks, excluding duplications.

different implications for each part. Figure 3, through the blue bars, shows that 52% of students did not include a duplicate implication across both assessment parts. The figure also shows that 43% of students had only one to three duplications. Only one student stood out, identifying eight duplicate implications, suggesting that only one student was able to identify and take advantage of the implicit feedback cycle. This suggests that possibility 2 is not at all likely.

One reason for this might have been that most students could not apply critical thinking skills to connect the two parts without explicit instruction. If the instructions clearly included an explicit step for students to take their findings in Part B and incorporate them back into Part A, it is highly likely that the number of duplications would have been much higher. This is not surprising, as implicit connections have been identified as something hard for students to identify (Kotluk and Tormey 2025). Another possibility is that the students simply choose not to act and reflect upon the feedback. As outlined in Nikolic, Stirling, and Ros (2018), even the best-intentioned feedback designs can do little for learning if the student decides not to act and reflect upon the feedback. Another alternative the authors believe may be most likely is that students may have simply been under the impression that they needed to find different implications across the two parts. As there was no post-assessment focus group, such analysis will require a follow-up study. The authors will use these insights to improve the assessment instructions for future use to maximise the learning outcomes.

As shown in Figure 3 through the orange bars, a large cluster of students identified 6–7 unique implications across both Parts A (median of 2) and B (median of 5), with an average of 6.6 and a median of 7. This is higher than each part separately, suggesting that the two approaches helped students identify more implications than one alone. Based on the greater number of implications identified through Part B the research approach appears to have been more effective.

On average, students could only identify 21% of the 32 possible implications. This suggests that effort must be made to explicitly encourage greater holistic student ethical awareness and that multiple approaches may be needed. The duplication of implications across Parts A & B was concentrated in only 14 codes. However, they were mainly concentrated across four implications: output quality and accuracy (C27, $n = 12$), bias (C4, $n = 7$), improved understanding (C31, $n = 7$) and

language fluency (C32, $n = 7$). This suggests that these are the implications students are most aware of and can most easily identify.

Identification of student awareness

By analysing the unique coding across Parts A and B, a general overview of student awareness can be obtained. It is important to identify where student awareness is high and where it is low, providing guidance as to where greater teaching and learning efforts on improving GenAI ethical awareness could be targeted. As demonstrated in Figure 4, student awareness was highest for improved understanding (C31), with 90% of students identifying this implication. This was followed by competitive advantage (C6) at 83%, output quality and accuracy (C27) at 63%, process automation (C3) and bias (C4) at 60%, language fluency (C32) at 58%, and human labour (C19) with 45%. Interestingly, the lecture content related to the assessment covered environmental impact (C15), human labour (C19) and ethical integrity (C20). Only C19 was identified by a large percentage of students. This suggests limited practical implementation of lecture content to aid student identification of the implications.

The orange bars in Figure 4 represent the implications that students should have been able to identify easily through Part A. As shown by the high percentage bars, these represent many of the most identified implications. This form of analysis suggests that Part A may have, in fact, helped students identify implications across both parts.

A recent study by the Quince, Petkoff et al. (2024), explored the considerations of most concern by academic staff. The greatest concern was for integrity, which the students did not highly identify; bias, which was the 4th highest identified implication by the students; and privacy and data security, which only received modest attention from the students. This suggests that there is mostly a mismatch between students and staff when identifying important ethical implications.

While Figure 4 provides a good understanding of the spectrum of implications identified by the students, it is important to understand how they recognised them. Figure 5 shows the frequency of the topics that were coded in both Parts A and B. As identified earlier, there are some contrasting differences between the implications students identified across each part.

Determine which implications need greater awareness by students

Table 2 provides a breakdown of the implications that were discussed by less than 5% (or less) of the cohort (3 students).

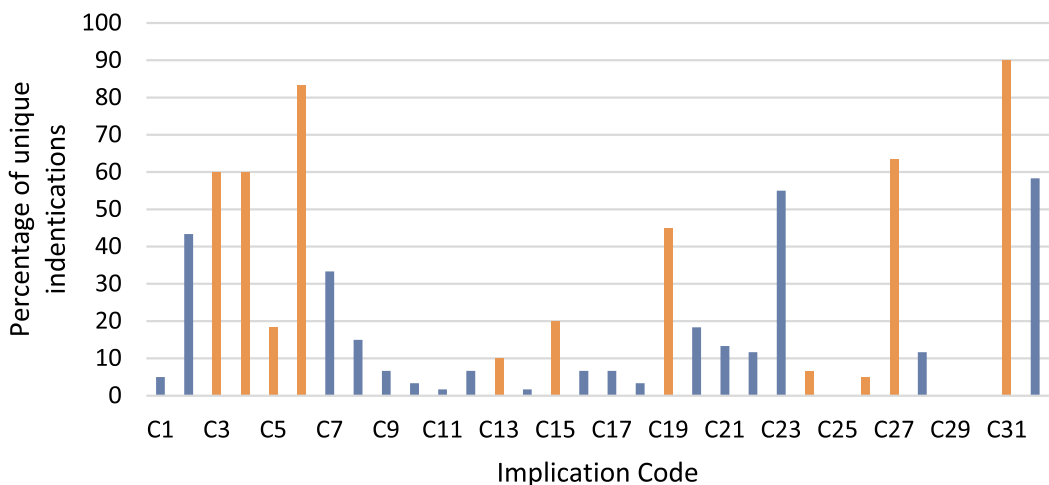


Figure 4. The percentage of unique identification for each implication code (orange bars represents codes associated with Part A).

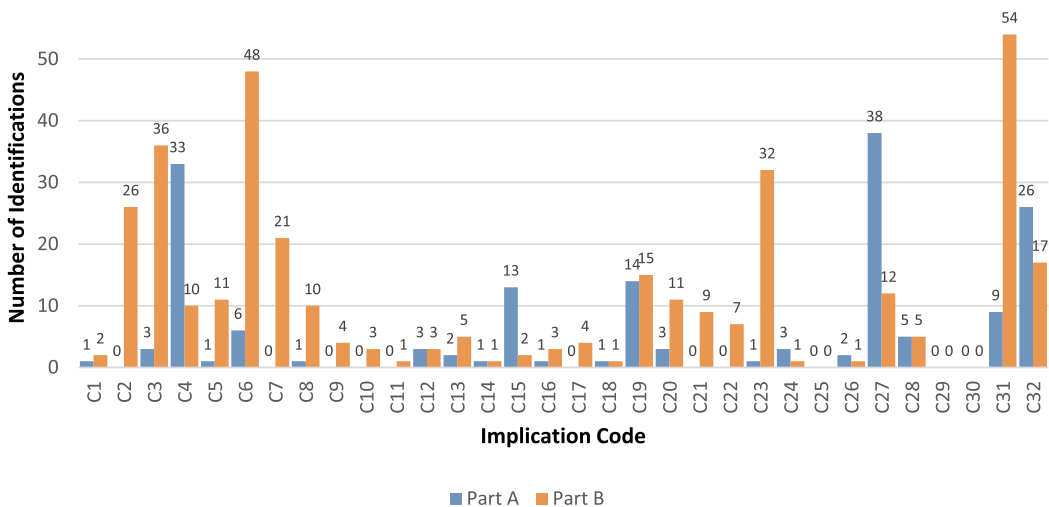


Figure 5. Frequency of topics coded in Parts A & B of the student’s assessments.

Based on the assessment design, ‘Affect recognition’ (C1, $n = 3$) and ‘Psychological Well-being’ (C26, $n = 3$) had implications that were not expected for students to raise. However, they were mentioned a few times around the use of input language and output language and the behaviours of the user being recognised. If GenAI can understand a user’s emotional and psychological well-being, GenAI can be used positively for improvement purposes, but in contrast, it can be used for substantial manipulation (Daniel and Nikolic 2023; Jung 2024). For this implication to be more commonly discussed, there may need to be a direct question around these terms.

‘Data collection and Utilisation’ (C10, $n = 2$), ‘Data Ownership’ (C11, $n = 1$), ‘Data Security’ (C12, $n = 4$), ‘Privacy’ (C25, $n = 0$) and ‘Safeguarding Against Misuse’ (C29, $n = 0$) are all implications that revolve around GenAI’s ability to track, relay and store information. While these topics were not explicitly guided to within the assessment design, they do play an important role in Part B’s focus on study and profession. Students need to be fairly aware of what information is uploaded and possibly used for training purposes, not just for themselves but also for the companies they may work for. Lack of awareness could lead to major financial and legal ramifications (Ferrara 2024). For these implications to be discussed, a direct question might be required to ask students to investigate the data handling and potential misuse of the information it gains.

‘Enhances user experiences’ (C14, $n = 1$) and ‘feedback and improvement’ (C18, $n = 2$) are interesting implications covering the ability of GenAI to create a better user environment and to provide

Table 2. Topics not covered by more than 5% of the cohort in either part of the assessment.

Affect recognition (C1)
Data collection and utilisation (C10)
Data ownership (C11)
Data security (C12)
Enhance user experiences (C14)
Equity and accessibility (C16)
Feedback and improvement (C18)
Power and hegemony (C24)
Privacy ^a (C25)
Psychological well-being (C26)
Safeguarding against misuse ^a (C29)
Transparency and accountability ^a (C30)

^aTopics that were not discussed in either part by any students.

useful learning feedback in multiple forms. With lifelong learning established as unavoidable for engineers (Guest 2006), this could impact a student's ability and motivation to learn, impacting the innovation and productivity gains available when in employment.

'Power and hegemony' (C24, $n = 4$) and 'equity and accessibility' (C16, $n = 4$) are topics that both work together to discuss fair training and use, as well as equitable availability of GenAI. These translate to the possible biased judgements that can be formed from GenAI use and the competitive advantages that can be gained by having access to specific GenAI's (Daniel and Nikolic 2023). With government policies and higher education institutions placing much emphasis on improving and raising awareness of fairness and equity (Cumming and Rose 2022; Salmi and D'Addio 2021), it is surprising that such focus did not gain greater attention. Again, a direct question would most likely be required to cover these topics.

'Transparency and accountability' (C30, $n = 0$) involves transparency in how GenAI systems operate, ensuring ethical use, addressing biases or errors, and being answerable for the impact and decisions made using AI technology. Whilst students did talk about the data that ChatGPT was trained on, they did not mention the transparency or accountability of the platform.

This analysis has shown the range of implications that the students struggled to identify. The question becomes, are these implications less important and deserving of being less known? Is there a need for a pedagogical redesign to ensure these implications gain greater attention? Such questions provide a pathway for future research.

Conclusions

This study addressed the research question, '*Using an embedded multi-stage assessment, how many social, economic and environmental ethical implications can students identify?*' where students learned by doing. The aim was to determine which implications students could identify using critical thinking. The findings show that from the 32 implementation types identified by the authors, 29 were identified by the students. The most identified implications were improved understanding (C31), competitive advantage (C6), output quality and accuracy (C27), process automation (C3) and bias (C4). This highlighted that students were more aware of the positive implications rather than the negative ones. Through this assessment design, privacy (C25), safeguard against misuse (C29), transparency and accountability (C30) were never raised in either part of the assessment. As most implications were identified by at least one student, and the remaining ones represent plausible possibilities, this suggests that the 32 Possible GenAI-based Social, Economic, and Environmental Implications and Uses are both valid and meaningful. Future studies can look at refinement.

Compared to some recent literature (Quince, Petkoff et al., 2024), a disconnect between important ethical implications between academic and student opinions seems evident. For Part A, students identified an average of 2.69 implications, with a median of 2. For Part B, students identified an average of 5.63 implications, and the median was 5. This suggests that the research component (Part B) was effective in helping students become more aware of the ethical implications compared to the direct evaluative judgment activity (Part A). Both parts combined resulted in an average of 6.6 and a median of 7 implication identifications. This suggests that using a multi-stage approach is beneficial as it resulted in the broadest engagement of implications, but this low value highlights the need to raise greater awareness.

The identification of implementations among students was diverse, so it shouldn't be assumed that students know particular ethical implications. This suggests that targeted activities are needed to raise engagement and awareness of GenAI's ethical implications. The data suggests that most students did not identify the implicit feedback cycle within the pedagogical design.

As this study was conducted in Q1 2023 when ChatGPT-3.5 usage in educational contexts was only gaining its feet, this study, even with the limited sample size, provides a unique benchmark of what students could identify in its early adoption. This assessment and study in this form cannot be repeated today due to the advancements in students' knowledge of the use of GenAI and the integrity risks

associated with unsupervised essays and assignments (Nikolic et al. 2024). There are other suggested learning-by-doing approaches to learning digital ethics, such as the Digital Ethics Canvas by Hardebolle et al. (2023) that could be considered but also limited as an unsupervised assessment. As considerable time has passed and GenAI literacy has developed, a future study would be interesting to identify whether students have advanced their understanding of GenAI implications.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data for this project is not publicly available.

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